

Mutual Fund Investor Onboarding & KYC Platform — Implementation Document

Overview

This document defines the implementation plan for a production-style mutual fund investor onboarding and KYC platform tailored for first-time Indian investors. The platform is designed not only to digitize onboarding, but also to simplify the process through beginner-friendly education, clear step guidance, strong trust signals, and visually guided investment discovery. KFinTech's investor-facing mutual fund services already expose digital KYC entry points such as e-KYC and Aadhaar KYC, which makes this problem statement suitable for a realistic industry-aligned implementation.[cite:2][cite:5][cite:8]

The product goal is to reduce user confusion during account creation, KYC completion, and first-time mutual fund investing. SEBI's investor education guidance states that KYC is a mandatory part of secure financial transactions for mutual funds, which means the onboarding flow should be designed as a compliance journey as well as a user education journey.[cite:9]

Problem Statement

Traditional onboarding flows in financial products often focus on form submission and compliance collection, but they do not sufficiently help first-time investors understand what mutual funds are, why KYC is needed, what each document means, and what happens after submission. This creates friction, incomplete applications, anxiety, and poor activation for new investors.

The implementation challenge is to build a digital onboarding system that is compliant in structure, simple in language, and visually intuitive enough that a first-time investor can understand the process step by step. The solution should help users move from registration to verified KYC status and then toward informed first investment decisions, while maintaining a trustworthy and modern product experience.[cite:2][cite:5][cite:9]

Product Vision

The platform should behave as an explainable onboarding experience rather than a generic signup form. It should guide users through identity creation, KYC, document verification, bank verification, risk understanding, and fund comparison in a way that is understandable even for beginners.

A suitable positioning statement is: an explainable mutual fund onboarding and KYC platform for first-time Indian investors that combines secure digital verification, plain-language education, visual fund comparison, and trust-building user experience. This direction aligns with existing KFinTech onboarding patterns and with the broader requirement that KYC be mandatory before secure financial participation.[cite:2][cite:5][cite:9]

Core Objectives

- Enable secure user registration and authentication.
- Support structured mutual fund onboarding from account creation to KYC completion.
- Explain each onboarding step in plain language for beginner investors.
- Simulate realistic KYC and document workflows used in Indian financial onboarding systems.
- Provide fund discovery and beginner-friendly comparison experiences after compliance completion.
- Offer admin-side visibility into verification, approval, rejection, resubmission, and audit tracking.
- Present all flows in a visually attractive, mobile-first, high-trust interface.

Target Users

Primary users

- First-time mutual fund investors in India.
- Users who know basic finance terms but are unfamiliar with the process.
- Younger digital users who prefer mobile-first onboarding.
- Users who need extra explanation before trusting financial forms and uploads.

Secondary users

- Operations and compliance reviewers.
- KYC verification staff.
- Admin users who track onboarding drop-offs and document issues.

Key Differentiators

Most onboarding products provide account creation, document upload, and status viewing. A better implementation should distinguish itself through onboarding clarity, financial education, and decision support. Competitor-style onboarding materials around Groww emphasize simplicity and staged activation, which suggests that a product can differentiate through better explanation quality and more guided investment readiness.[cite:4][cite:7]

The proposed differentiators are:

- Beginner mode with simple explanations for PAN, Aadhaar, SIP, folio, NAV, risk, returns, and lock-in.
- “Why this is needed” hints beside every important data field.
- Visual progress tracker from signup to first investment readiness.
- OCR-assisted or simulated document extraction preview for user confidence.

- Fund comparison cards written in plain language rather than only numerical tables.
- Trust center showing KYC stage, pending actions, document quality feedback, and notification history.
- Guided first-investment simulation that explains impact without making promises of returns.

Scope

In scope

- Investor registration and login.
- OTP-based verification flow simulation.
- PAN and Aadhaar-based KYC intake flow.
- Document upload and review journey.
- KYC status tracking.
- Bank account verification simulation.
- Fund discovery and comparison UI.
- Beginner education content modules.
- Admin dashboard for review actions and status control.
- Audit trail and notification center.

Out of scope

- Real payment gateway settlement.
- Real external Aadhaar or PAN verification APIs in the student version.
- Actual investment execution into live mutual fund schemes.
- Regulatory production certification.
- Real bank penny-drop integration unless mocked.

Functional Modules

1. Investor Authentication Module

This module is responsible for initial access control and identity-linked account creation. Authentication should be separate from KYC completion because a user may be registered but still not be eligible to invest until compliance steps are complete.[cite:2][cite:9]

Features

- Sign up with mobile number or email.
- OTP verification flow.
- Secure login with JWT-based session handling.
- Password or MPIN setup for return login.
- Forgot password or OTP recovery flow.
- Device/session management for future enhancement.

User states

- Registered but not verified.
- Verified but KYC not started.
- KYC in progress.
- KYC submitted.
- KYC approved.
- KYC rejected or resubmission required.
- Investment-ready.

2. Profile and Identity Module

This module collects personal identity details required to create the investor profile. KFintech's own investor services reference Aadhaar and PAN based KYC journeys, so this module should be modeled as a structured intake stage before full verification.[cite:5][cite:8]

Features

- Basic profile creation.
- PAN input and validation format checks.
- Aadhaar-linked KYC choice flow.
- Date of birth, contact details, address, and occupation fields.
- Nominee and optional guardian details.
- Profile completion score.

3. KYC Workflow Module

This is the central module of the platform. SEBI identifies KYC as mandatory for mutual fund participation, and KFintech offers digital KYC process entry points, so the implementation should simulate a realistic verification chain.[cite:2][cite:5][cite:9]

Features

- KYC start screen with explanation of why KYC is needed.
- Document checklist.
- PAN card upload.
- Aadhaar or address proof upload.
- Selfie or live-photo capture simulation.
- OCR extraction simulation for PAN details.
- Document quality validation messages.
- KYC submission summary.
- Real-time status updates.
- Reupload workflow for rejected documents.

KYC statuses

- Not started.
- In progress.
- Pending documents.
- Under review.
- Approved.
- Rejected.
- Resubmission requested.

4. Bank Verification Module

Bank verification is necessary before moving toward transaction readiness. The platform can simulate this stage through structured form validation and status messaging rather than live banking integration.

Features

- Bank account holder name.
- Account number and IFSC collection.
- Cancelled cheque or passbook upload.
- Verification result simulation.
- Status messages for mismatch and retry.

5. Investor Education Module

This module is the main innovation layer. The user requirement calls for a system that even a child could understand, which should be interpreted as making complex financial steps visually simple, plain-language, and guided.

Features

- “Explain simply” mode for finance terms.
- Tooltip-based explanation for each field.
- Mini cards for concepts such as SIP, lump sum, NAV, folio, KYC, and risk.
- Step-end explanation: what was completed, why it mattered, and what happens next.
- Interactive glossary.
- Beginner checklists before first investment.

6. Fund Discovery and Comparison Module

After KYC approval, users should not be dropped into a raw data list. Instead, they should see a guided exploration experience. The comparison layer should help users compare funds on understandable attributes like category, risk, minimum amount, and suitability.

Features

- Search by category.
- Beginner-focused tags such as “Low volatility”, “Tax saving”, or “Long-term wealth”.
- Filter by minimum investment amount and risk level.
- Side-by-side comparison cards.
- Graphs for educational illustration only.
- Save to watchlist.
- “Good for whom” and “not ideal for whom” descriptions.

7. Notification and Trust Module

Trust is critical in fintech onboarding. Users should always know what was submitted, what is pending, and whether action is required. KFinTech’s investor flows expose KYC status-related interactions, which supports the inclusion of visible status communication.[cite:2][cite:8]

Features

- Status updates for KYC stage changes.
- In-app notifications.
- Email or SMS simulation hooks.
- Submission receipts.

- Audit trail timeline.
- Security notices for failed login attempts.

8. Admin and Review Module

This module supports operational users responsible for document review and issue handling.

Features

- Admin login.
- Investor list with filters by status.
- KYC document review screen.
- Accept, reject, request resubmission.
- Review comments and reason tagging.
- Audit logs.
- Basic analytics for drop-offs and approval rates.

User Journey

Investor flow

1. User lands on beginner-friendly homepage.
2. User creates account with mobile or email.
3. OTP verification is completed.
4. User fills basic profile details.
5. User starts KYC after seeing a short explanation.
6. User uploads PAN and address/Aadhaar documents.
7. System shows extracted details and asks for confirmation.
8. User uploads bank proof.
9. KYC is submitted and status becomes under review.
10. If approved, user enters fund comparison and first-investment education screens.
11. If rejected, user sees specific reasons and resubmits.

Admin flow

1. Admin logs in.
2. Admin sees applications grouped by status.
3. Admin opens investor record.
4. Admin reviews uploaded data and document previews.
5. Admin approves, rejects, or asks for resubmission.

6. System logs action and notifies investor.

Information Architecture

The platform should be structured around user clarity rather than technical system grouping.

Investor-side navigation

- Home
- Start onboarding
- My profile
- KYC center
- Documents
- Bank verification
- Learn
- Compare funds
- Notifications
- Help and support

Admin-side navigation

- Dashboard
- Investor applications
- KYC review queue
- Rejected cases
- Audit logs
- Reports
- Settings

System Architecture

A modular three-tier architecture is appropriate for this implementation.

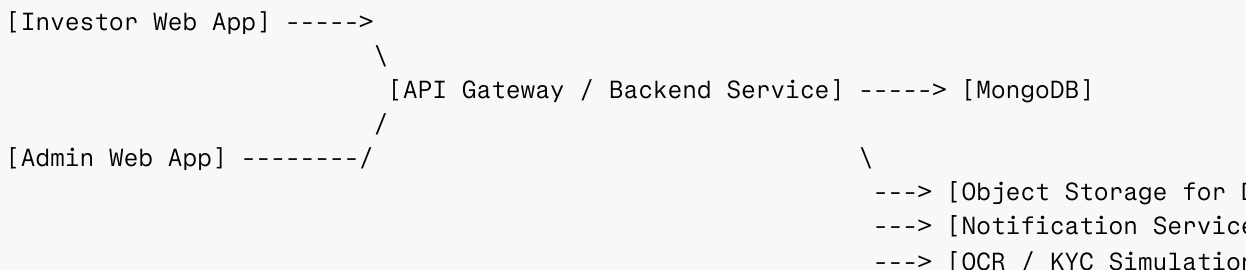
Architecture layers

Layer	Responsibility	Suggested Technology
Presentation layer	User-facing investor UI and admin UI	React or Next.js
Application layer	Business logic, auth, workflows, validation, notifications	Node.js with Express
Data layer	User data, KYC records, fund data, audit logs	MongoDB
File storage	Uploaded documents and images	AWS S3 or S3-compatible bucket

Layer	Responsibility	Suggested Technology
Deployment layer	Containerization and hosting	Docker

This architecture matches the project direction shown in the original problem statement image, which lists React, Node.js with Express, MongoDB, AWS S3, and Docker as the core stack. [cite:2]

High-Level Architecture Diagram



Frontend Architecture

The frontend should follow component-driven design with role-based route separation.

Investor app structure

- Authentication pages.
- Onboarding dashboard.
- KYC wizard.
- Document center.
- Education center.
- Fund comparison pages.
- Status and notification pages.

Admin app structure

- Login.
- Review dashboard.
- Application detail screen.
- Queue management.
- Analytics summary.

Frontend design principles

- Mobile-first responsive design.
- Large clear progress indicators.
- Friendly empty states.
- Low-jargon content.
- High-contrast trust-focused color palette.
- Distinct states for success, pending, rejected, and action required.
- Visual hierarchy that explains process before asking for action.

Backend Architecture

The backend should expose REST APIs grouped by domain modules.

Suggested services

- Auth service.
- User profile service.
- KYC service.
- Document service.
- Bank verification service.
- Fund catalog service.
- Notification service.
- Admin review service.
- Audit service.

Example API groups

- `/api/auth/*`
- `/api/users/*`
- `/api/kyc/*`
- `/api/documents/*`
- `/api/bank/*`
- `/api/funds/*`
- `/api/notifications/*`
- `/api/admin/*`
- `/api/audit/*`

Data Architecture

MongoDB is suitable for this student project because onboarding workflows include flexible user profiles, document metadata, notification records, and fund comparison data. The data model should separate investor identity data, workflow status, and uploaded asset references.

Core collections

- users
- investor_profiles
- kyc_records
- documents
- bank_accounts
- funds
- watchlists
- notifications
- audit_logs
- admin_users

Key relationships

- One user has one investor profile.
- One user has one or many document records.
- One user has one KYC record with status history.
- One admin reviews many KYC records.
- One user can save many funds to watchlist.

Suggested Database Schema

users

- `_id`
- `fullName`
- `email`
- `mobile`
- `passwordHash`
- `role` (investor or admin)
- `isVerified`
- `createdAt`
- `updatedAt`

investor_profiles

- `userId`
- `panNumber`
- `aadhaarMasked`
- `dob`
- `gender`
- `occupation`
- `annualIncomeRange`
- `address`
- `nominee`
- `profileCompletion`

kyc_records

- `userId`
- `status`
- `currentStep`
- `submittedAt`
- `reviewedAt`
- `reviewedBy`
- `rejectionReasons`
- `statusHistory`

documents

- `userId`
- `documentType`
- `fileUrl`
- `storageKey`
- `ocrExtractedData`
- `verificationStatus`
- `uploadedAt`

bank_accounts

- `userId`
- `accountHolderName`
- `accountNumberMasked`

- ifsc
- verificationStatus
- proofDocumentUrl

funds

- schemeName
- category
- riskLevel
- expenseRatio
- minimumInvestment
- description
- tags

notifications

- userId
- type
- title
- message
- isRead
- createdAt

audit_logs

- actorId
- actorRole
- action
- entityType
- entityId
- metadata
- timestamp

Authentication and Authorization Architecture

Authentication should be implemented using JWT for stateless session handling in the student build. The system should issue access tokens after OTP verification or credential validation, and protected routes should validate both identity and role before granting access.

Recommended approach

- OTP verification during registration.
- Password hash storage using bcrypt.
- JWT access token for API access.
- Refresh token strategy if session persistence is needed.
- Role-based access control for investor and admin routes.
- Middleware to restrict admin-only review endpoints.

Authorization model

Role	Permissions
Investor	Register, login, update profile, upload documents, view KYC status, compare funds
Admin	View all applications, review documents, change KYC status, add comments, access reports

KYC Workflow Architecture

The KYC engine should behave like a state machine so that transitions are predictable and auditable.

State transitions

```
Not Started -> In Progress -> Pending Documents -> Submitted -> Under Review -> Approved
                                                    |
                                                    |
                                                    -> Rejected
                                                    -> Resubmission Requested -> Submitted
```

Workflow logic

- The system validates whether mandatory identity fields are complete.
- The system validates document presence and basic file quality.
- OCR extraction is run or simulated for PAN validation support.
- Admin review determines approval, rejection, or resubmission.
- Each status change is written to status history and audit logs.
- Each critical change triggers a user notification.

Document Management Architecture

All user-uploaded documents should be stored outside the main database in object storage, with only metadata and storage references stored in MongoDB. This reduces database size and supports scalable file handling.

Document flow

1. User uploads file.
2. Backend validates type and size.
3. File is stored in S3 bucket.
4. Metadata is stored in documents collection.
5. OCR simulation processes file if applicable.
6. Admin accesses secure preview through backend.

Document types

- PAN card.
- Aadhaar or address proof.
- Selfie or live image.
- Cancelled cheque or passbook.
- Signature image if needed.

UI/UX Architecture

The platform should be designed around trust, explanation, and visual clarity.

Core UX principles

- Show one primary action at a time.
- Always explain why a document or field is required.
- Use visual step progress with percentage or stage markers.
- Avoid banking jargon without explanation.
- Show real-time validation feedback.
- Use icons, illustrations, and cards to reduce text heaviness.
- Keep mobile usability equal to desktop usability.

Key screens

- Landing page with clear onboarding CTA.
- Signup/login screen.
- Onboarding dashboard.
- KYC wizard screen.
- Document preview and confirmation screen.
- KYC status tracker.
- Compare funds screen.

- Learn mutual funds screen.
- Admin review dashboard.

Competitor-Informed Design Direction

Existing digital investment products in India emphasize fast activation and clean onboarding, but there is still room to improve explainability and trust-led financial education. KFinTech's investor services provide KYC-centered access points, while competitor onboarding references around Groww highlight the importance of simplicity and progressive steps.[cite:2][cite:4][cite:7][cite:8]

Proposed UI positioning

- More educational than generic investment apps.
- More visual than standard compliance portals.
- More trustworthy than flashy fintech landing pages.
- More beginner-focused than expert-first dashboards.

Non-Functional Requirements

Security

- Password hashing.
- JWT verification middleware.
- Rate limiting for OTP and login endpoints.
- Secure upload validation.
- Role-based access control.
- Audit logging for sensitive actions.
- Masked display for Aadhaar and bank numbers.

Performance

- Lazy loading for document previews.
- Pagination for admin queues.
- Optimized image compression.
- CDN-backed document delivery where applicable.

Scalability

- Modular services for auth, KYC, and notifications.
- External object storage for files.
- Containerized deployment with Docker.

Accessibility

- Clear focus states.
- Readable font sizes.
- High contrast labels.
- Simple language mode.
- Mobile touch-friendly actions.

Suggested Tech Stack

Layer	Technology
Frontend	React or Next.js
Backend	Node.js + Express
Database	MongoDB
Storage	AWS S3
Authentication	JWT + bcrypt + OTP service
Containerization	Docker
Styling	Tailwind CSS or modular CSS
Charts	Recharts or Chart.js

This stack remains aligned with the original project framing and is appropriate for a portfolio-ready implementation.[cite:2]

API Design Outline

Authentication

- POST /api/auth/register
- POST /api/auth/verify-otp
- POST /api/auth/login
- POST /api/auth/forgot-password

User profile

- GET /api/users/me
- PUT /api/users/me

KYC

- POST /api/kyc/start
- POST /api/kyc/submit
- GET /api/kyc/status
- PUT /api/kyc/resubmit

Documents

- POST /api/documents/upload
- GET /api/documents/:id
- DELETE /api/documents/:id

Bank verification

- POST /api/bank/verify
- GET /api/bank/status

Funds

- GET /api/funds
- GET /api/funds/:id
- POST /api/funds/compare
- POST /api/funds/watchlist

Notifications

- GET /api/notifications
- PUT /api/notifications/:id/read

Admin

- GET /api/admin/applications
- GET /api/admin/applications/:id
- PUT /api/admin/applications/:id/review
- GET /api/admin/reports

Deployment Architecture

A simple deployment model can be used for the student version.

Deployment plan

- Frontend hosted on Vercel or Netlify.
- Backend hosted on Render, Railway, or AWS.
- MongoDB Atlas for managed database hosting.
- AWS S3 for documents.
- Docker used for consistent local and deployment environments.

Development Roadmap

Phase 1: Foundation

- Set up frontend and backend repositories.
- Configure authentication.
- Build base layout and routing.
- Create MongoDB models.

Phase 2: KYC core

- Build profile flow.
- Build KYC wizard.
- Add document upload.
- Add status state machine.

Phase 3: Admin and review

- Build admin dashboard.
- Add review actions.
- Add audit trail and notifications.

Phase 4: Differentiation

- Add beginner education center.
- Add fund comparison module.
- Add visual trust center and progress insights.

Phase 5: Polish

- Improve UI responsiveness.
- Add analytics and validation improvements.
- Improve empty states, tooltips, and onboarding copy.

Risks and Mitigations

Risk	Impact	Mitigation
Generic implementation looks like a simple CRUD app	Low project impact and weak differentiation	Focus on guided education and strong onboarding UX
Too many compliance flows make UX complex	User drop-off	Use progressive disclosure and step-by-step wizard
No live verification APIs available	Reduced realism	Simulate external services and clearly document mock boundaries
Poor document upload experience	Failed onboarding completion	Add preview, quality hints, and clear retry flow
Weak admin visibility	Hard to review cases	Build review queue, filters, and audit logs

Success Metrics

The project can be evaluated using product and UX metrics even in a prototype.

Suggested metrics

- Registration completion rate.
- KYC start rate.
- KYC completion rate.
- Document rejection rate.
- Average time to submit KYC.
- Fund comparison engagement rate.
- Beginner help content usage rate.
- Resubmission success rate.

Conclusion

This implementation should be presented not as a basic onboarding form, but as a beginner-first, explainable mutual fund onboarding and KYC platform. The strongest version of the project combines secure authentication, realistic workflow architecture, document handling, admin review, guided education, and visually clear comparison experiences. KFinTech's public investor flows and SEBI's KYC guidance make this a credible, industry-aligned product direction, while the educational and UI-led enhancements create a stronger and more interesting final solution.[cite:2][cite:5][cite:8][cite:9]